

A Lightweight Hybrid Deep Learning Framework for Image-Based Air Quality Index Classification

Abstract

Air pollution poses significant health risks globally, necessitating continuous and scalable monitoring solutions. Traditional sensor-based monitoring stations suffer from high deployment costs and limited spatial coverage. To address these limitations, image-based Air Quality Index (AQI) classification has emerged as a promising alternative. However, existing deep learning frameworks either rely on heavy backbones with substantial computational latency or require extra multimodal sensor hardware. In this paper, we propose a lightweight hybrid architecture designed for multi-class AQI classification using ground-level sky imagery alone. The model combines a frozen pre-trained ResNet50 backbone, 1×1 channel reduction, bilinear upsampling, spatial patchification into a 4×4 grid, and a shared Cross Stage Partial Block (CSPBlock). Evaluated on a benchmark dataset of 12,240 images across six AQI severity categories, it achieves a classification accuracy of 96.73% while maintaining a low inference latency of 4.400 ms per image, delivering a favorable balance between accuracy and computational efficiency for real-time edge deployment.

Index Terms

Air Quality Index (AQI), Deep Learning, Cross Stage Partial Networks, Image Classification, Lightweight Neural Networks, Atmospheric Haze.

I. INTRODUCTION

A. Background Study

According to the World Health Organization (WHO), air pollution is considered to be one of the major risks to child health. From the data of 2019, it is estimated that almost 99% of world population lives in poor air quality. Such air quality damages human respiratory functions resulting in almost 4.2 million premature deaths worldwide in 2019 [1].

According to a recent study done in Bangladesh, air pollution is considered as a major contributing factor in respiratory illness. It resulted in asthma, COPD and other respiratory illnesses. The study mentioned factors like industrial pollutants, dust from construction sites and combustion of fuel as the primary source of air pollution [2].

AQI is a system that tracks the concentration of ozone, particulate matter and other air pollutants such as carbon monoxide (CO), sulphur dioxide (SO₂) etc. Depending on the concentration of these air pollutants, AQI is divided into six categories and each category is given a range which is called the AQI index. The six categories are Good (0–50), Moderate (51–100), Unhealthy for Sensitive Groups (101–150), Unhealthy (151–200), Very Unhealthy (201–300), and Hazardous (301–500) [3].

TABLE I
AIR QUALITY INDEX LEVELS AND HEALTH CONCERN CATEGORIES [4]

AQI Category	AQI Range	Health Impact / Meaning
Good	0–50	Air quality is considered satisfactory, and air pollution poses little or no risk.
Moderate	51–100	Air quality is acceptable; however, sensitive individuals may experience minor health concerns.
Unhealthy for Sensitive Groups	101–150	Members of sensitive groups may experience health effects; general public is unlikely to be affected.
Unhealthy	151–200	Everyone may begin to experience health effects; sensitive groups may experience more serious effects.
Very Unhealthy	201–300	Health warnings of emergency conditions. The entire population is more likely to be affected.
Hazardous	301–500	Health alert: everyone may experience more serious health effects.

Table I outlines the six standard Air Quality Index categories, their numerical boundaries, and their corresponding health concern levels [4].

Particulate Matter refers to a complex blend of solid and liquid particles suspended in the air, categorized by its diameter for regulatory air quality consideration. PM 10 indicates the particle size is 10 micrometers whereas PM 2.5 are the fine particles that are 2.5 micrometres or smaller in diameter [5]. Among these particles, PM 2.5 poses significant risks to human health because it can deeply infiltrate the lungs [6]. Experimental findings indicate that PM2.5 might possess greater cytotoxicity and cause more damage to humans via oxidative stress compared to PM10 [7]. In Bangladesh specifically, exposure to elevated particulate matter has been linked to increased respiratory and cardiovascular health risks, with vulnerable groups such as infants, schoolchildren, and the elderly facing disproportionately higher susceptibility to these effects [8]. Both pollutants act as a key indicator for calculating AQI. Their measured concentration in micrograms per cubic meter are converted into sub-index value, with the highest sub-index determining the overall AQI category [9].

AQI is traditionally monitored and measured by government operated stationary weather stations. While these stations provide accurate data, they are often expensive to deploy and maintain [10]. Due to the limited number of AQI monitoring stations, its coverage area is restricted, preventing it from delivering timely and high spatial resolution measurements effectively [10].

B. Literature Review

As conventional sensor-based AQI monitoring systems have high deployment cost and limited spatial coverage, it motivated researchers to explore image-based alternatives for air quality assessments. Since smartphones, surveillance networks, and satellite platforms use cameras extensively, image processing has become a promising option for scalable and cost-effective environmental monitoring. Particles in the atmosphere scatter light depending on their wavelength; it affects the color, contrast, and clarity of sky images. These are the elements that provide quantifiable data regarding pollution levels.

With recent advancements of machine learning and deep learning in computer vision, air quality can be assessed accurately from images by extracting meaningful patterns. By understanding the relation between visual characteristics such as color distortion, texture variation, and scene visibility with associated AQI levels, these models can accurately categorize air quality levels. Deep learning architectures, especially convolutional neural networks, further improve this by automatically learning hierarchical feature representations directly from raw images.

This review discusses three main categories of image-based AQI classification methods: traditional image processing methods, machine learning approaches, and deep learning frameworks, following the evolution from manually crafted physical models to fully learned representations.

1) *Traditional Image Processing Methods:* In early days estimating air quality depended on manually crafted physical and statistical attributes derived from atmospheric scattering models. The fundamental idea is that pollutants like PM 2.5 and PM 10 lead to quantifiable reductions in image contrast, brightness, color intensity, and visibility.

An earlier study [11] demonstrated this principle by collecting manually crafted features like luminance, brightness histogram data, and color adjustment values from a sky image. After feeding all these data into K-nearest neighbor classifier, it gave 92% accuracy in distinguishing hazy from clear conditions.

Expanding on this, Chu et al. [12] introduced fastDBCP, which measured the ratio of dark to bright channel prior information in down-sampled images as a specific visual pollution index. This approach showed a strong image-size-invariant relationship with AQI indicators such as PM 2.5 and PM 10 with more accuracy and processing speed.

Although these physically-grounded methods showed that visual image features contain significant data regarding air pollution levels, their performance degrades in different environmental circumstances. Due to dependence on basic atmospheric models and manually crafted features, researchers explored data-driven machine learning methods that can learn more detailed features from images automatically.

2) *Machine Learning Methods:* To address these limitations, researchers turned to supervised machine learning methods which can learn the discriminative relationship between image features and AQI levels using labelled data. This method extracts manual features from sky images and feeds them into trained ML classifiers to associate them with respective pollution categories.

W. Wang et al. [13] used this method by extracting color moments, color correlogram, and wavelet features from sky images and feeding them into a Support Vector Machine classifier to predict AQI. It achieved 82.3% accuracy on a dataset of 1,500 images.

Building on this concept, Chuchro et al. [14] implemented texture analysis on daily sky images taken in a city of Poland, averaging characteristics from five video frames before training both a regression and classification model for estimating PM 2.5 and PM 10. The classification model achieved accuracies of 0.86 and 0.80 for PM10 and PM 2.5 respectively. On the other hand, regression models gave R^2 values of 0.85 for PM10 and 0.63 for PM 2.5, showing that enhanced texture-based representations can increase estimation precision for various pollutant types.

Despite these advanced features in machine learning, it remains limited due to its dependency on manual feature engineering. These created manual features did not generalize well across various locations, lighting conditions, and camera types. Due to this limitation, researchers started using deep learning, which provided end-to-end hierarchical feature learning straight from raw images.

3) *Deep Learning Methods:* The emergence of deep learning, especially convolutional neural networks, has fundamentally changed image-based AQI classification by eliminating manual input feature engineering by enabling hierarchical feature learning from raw images.

A study [15] compared two CNN frameworks, VGG19 and ResNet50, for air quality classification based on environmental images classified into three groups: good, moderate, and severe. ResNet50 surpassed VGG19, attaining 97.87% accuracy, as its residual connections facilitated deeper feature learning without loss.

Acknowledging the impracticality of large CNN models for real-world use on mobile and edge devices, a multiscale dilated architecture [16] rearranged dilated convolutional filters to capture global and semantically higher-level features at the same time. It decreased parameters by 86.7% and floating-point operations by 88.5% in comparison to leading models, while preserving a top-1 classification accuracy of 94.2%, showing that lightweight CNN designs can reach competitive accuracy without computational demands of conventional architectures.

4) *Multimodal Deep Learning Approaches:* Recent advancements in multimodal deep learning frameworks showed excellent results where combining additional weather data from sensors with visual input can enhance classification accuracy.

AQFusionNet [17] implemented this multimodal strategy by combining ground-level atmospheric imagery with pollutant concentration data and training it with CNN architectures—MobileNetV2, ResNet18, and EfficientNet-B0—using semantically aligned embedding spaces for feature integration. EfficientNet-B0 attained the highest performance with a classification accuracy of 92.02% and an RMSE of 7.70 using over 8,000 samples from India and Nepal. It showed 18.5% improvement over single-modality baselines with low computation.

Expanding on this, Vision-AQ [18] introduced a dual-input architecture that combines pretrained ResNet50 for extracting image features with a multi-layer perceptron handling tabular sensor data of PM 2.5, PM 10, and AQI values. It achieved 99% accuracy in predicting six AQI categories. Moreover, it incorporated Grad-CAM Visualization to prove that the model relied on atmospheric characteristics like haze and smog patterns.

The most advanced framework reviewed here utilized a two-phase CapsuleNet-based structure [19], where the image-only phase reached 98.22% accuracy and full multimodal fusion achieved 99.98% accuracy, along with a precision of 99.86% and an F1-score of 99.61%.

Despite recent progress, current image-based AQI models have clear drawbacks. Traditional machine learning methods rely on manual features that do not work well under changing weather conditions, while deep learning and multimodal models are computationally heavy or need extra weather sensors, making them hard to run on edge devices. To solve this, this study introduces a lightweight hybrid deep learning model for multi-class AQI classification using only single-modality sky images. By pairing a frozen pre-trained feature extractor with spatial patch division and lightweight feature aggregation, the model captures fine haze details without extra hardware, achieving strong accuracy and fast inference speed for edge deployment.

II. MATERIALS AND METHODS

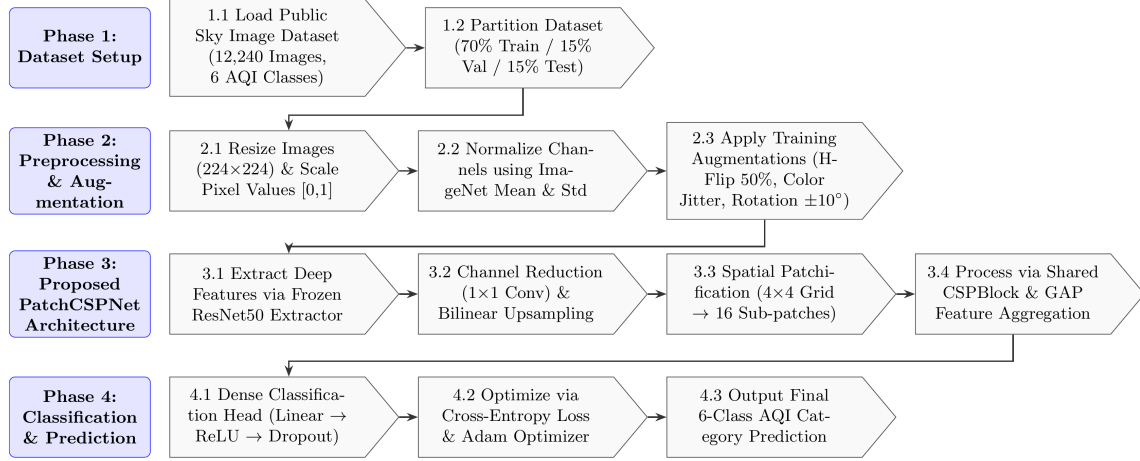


Fig. 1. End-to-end methodology pipeline for AQI classification

A. Dataset Description and Exploration

To evaluate the proposed architecture, this study utilizes the Air Pollution Image Dataset from India and Nepal, a publicly available benchmark repository [20]. The dataset consists of a total of 12,240 ground-level outdoor sky images captured across diverse urban and semi-urban geographic regions of India and Nepal under varying meteorological conditions. All native images possess an identical spatial resolution of 224×224 pixels, formatted as three-channel RGB matrices. The collection is strictly partitioned into six distinct categories, each mapped directly to the numerical ranges and corresponding levels of health concern defined by the standard Air Quality Index (AQI).

TABLE II
DATASET CATEGORIES AND CLASS DISTRIBUTIONS

Class Label	AQI Numerical Range	Image Count
a_Good	0–50	1,541
b_Moderate	51–100	1,573
c_Unhealthy_for_Sensitive_Groups	101–150	2,863
d_Unhealthy	151–200	2,622
e_Very_Unhealthy	201–300	2,194
f_Severe	301–500	1,447
Total	0–500	12,240

Table II provides a comprehensive structural breakdown of the dataset, detailing the class labels, operational AQI boundaries, and images per class.

Note: The `f_Severe` label corresponds to the *Hazardous* category (301–500) defined in Table I.

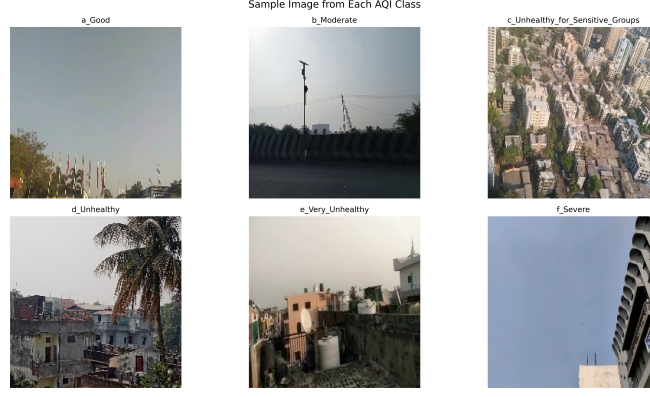


Fig. 2. Ground-level sky image samples from each of the six classification categories.

Figure 2 illustrates a representative image sample from each of the six classification folders, revealing the progressive visual degradation, color distortions, and structural occlusion of distant background targets caused by escalating ambient particulate matter.

To establish a rigorous evaluation pipeline, the dataset was split into an independent 70% training partition, a 15% validation partition, and a 15% testing partition, as detailed in Table III.

TABLE III
EXPERIMENTAL DATA PARTITIONING

Class Label	Train (70%)	Val (15%)	Test (15%)
a_Good	1,091	219	231
b_Moderate	1,113	229	231
c_Unhealthy_Sensitive	2,009	413	441
d_Unhealthy	1,837	400	385
e_Very_Unhealthy	1,520	337	337
f_Severe	998	238	211
Total	8,568	1,836	1,836

B. Data Preprocessing and Augmentation

To ensure structural compatibility across network layers and stabilize model convergence, a multi-stage image processing pipeline was established. The operations are divided into deterministic preprocessing applied identically across all dataset splits and stochastic data augmentations, which are applied exclusively to the training set to prevent data leakage and preserve evaluation integrity.

1) *Preprocessing Pipeline*: All input images are processed through a uniform pipeline to standardize structural dimensions and pixel value distributions prior to model ingestion:

- **Spatial Resizing**: Input images are uniformly resized to a spatial dimension of 224×224 pixels, establishing a consistent tensor input shape across the entire network.
- **Tensor Conversion and Range Scaling**: Pixel intensities are converted to single-precision floating-point tensors, automatically scaling raw color values from the integer range $[0, 255]$ to a normalized real-valued interval of $[0, 1]$.
- **Channel Distribution Normalization**: To prevent early gradient instability and accelerate optimization convergence, channel intensity distributions are standardized using ImageNet mean ($\mu = [0.485, 0.456, 0.406]$) and standard deviation ($\sigma = [0.229, 0.224, 0.225]$) statistics.

2) *Training Augmentation*: To expand training variety artificially and improve model generalization under varying environmental conditions, three controlled transformations are introduced into the training split:

- **Horizontal Flipping**: Images are horizontally flipped with a 50% probability to introduce spatial variation without distorting class-specific pollution indicators.

- **Color Distortion:** Random radiometric perturbations are applied to brightness ($\pm 30\%$), contrast ($\pm 30\%$), saturation ($\pm 30\%$), and hue ($\pm 10\%$) to prevent reliance on camera white-balance or solar angles.
- **Rotational Perturbations:** Images are randomly rotated within a range of $\pm 10^\circ$ to model camera tilts from handheld devices or sensor oscillations.

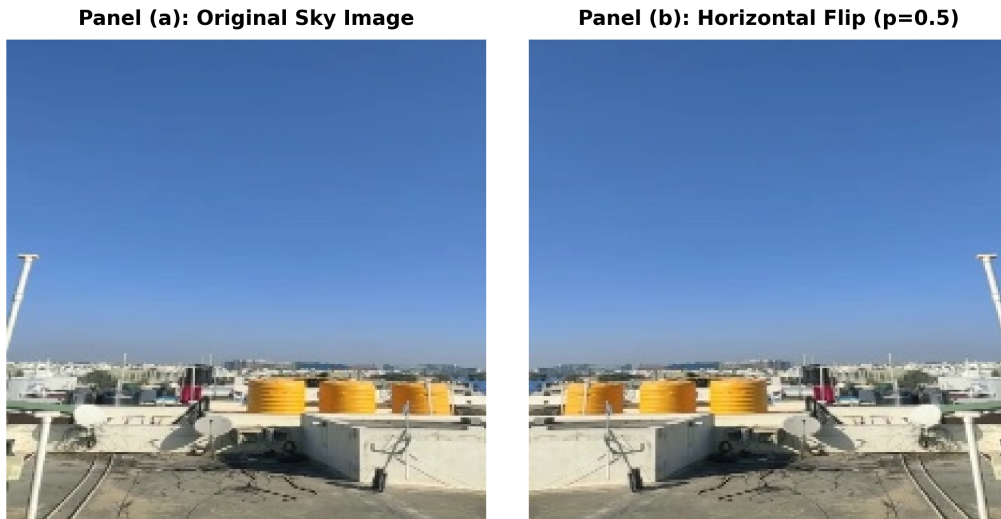


Fig. 3. Comparison of original sky image and augmented sample subjected to stochastic horizontal flipping ($p = 0.5$) to expand spatial variation.



Fig. 4. Comparison of original sky image and augmented sample subjected to radiometric perturbations (Color Jitter: brightness, contrast, saturation $\pm 30\%$, hue $\pm 10\%$) to model varying solar angles and lighting conditions.

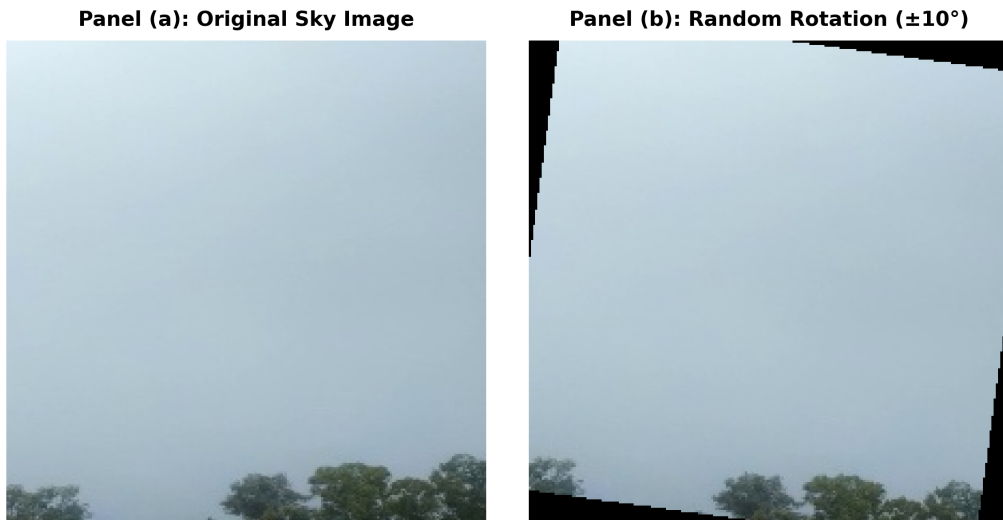


Fig. 5. Comparison of original sky image and augmented sample subjected to rotational transformations ($\pm 10^\circ$ Random Rotation) to simulate minor camera tilts.

C. Proposed Architecture

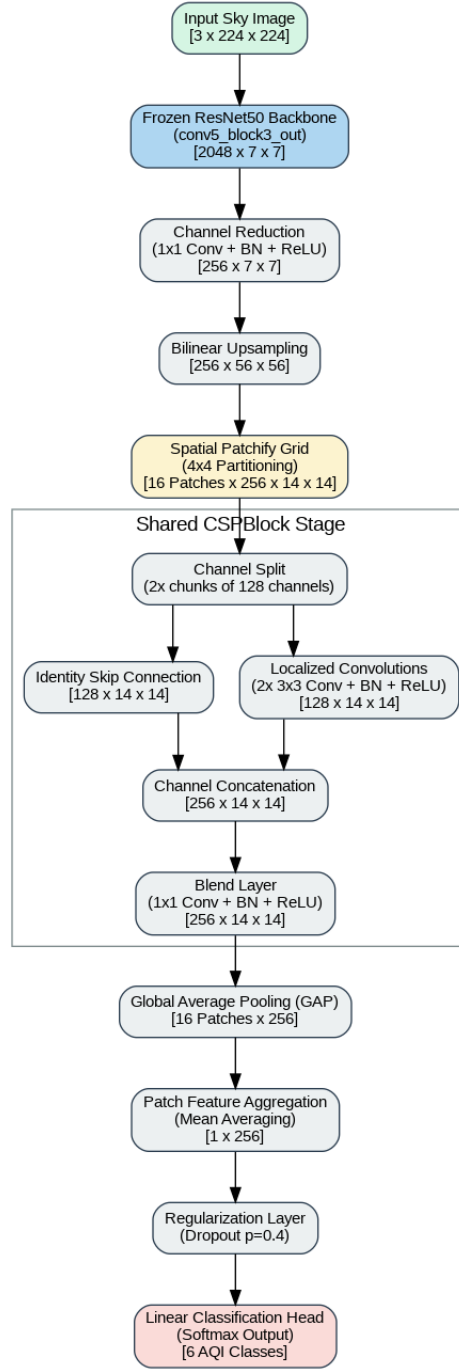


Fig. 6. End-to-end network architecture for AQI classification

To accurately determine air pollution levels from ground level sky photographs while keeping the system computationally lightweight, this study introduces a hybrid deep learning model. The workflow operates through four main stages:

1) *Pre-trained Feature Extraction*: The first stage extracts high-level visual representations from the input sky photograph using a ResNet50 backbone pre-trained on ImageNet. ResNet50 was selected for its residual learning framework, which addresses the vanishing-gradient and degradation problems that arise when training very deep convolutional networks [21]. Rather than requiring each stacked block to learn a full underlying mapping $H(x)$

directly, residual blocks reformulate the learning objective as a residual function relative to the block input:

$$y = F(x) + x \quad (1)$$

where x is the block input, $F(x)$ represents the learned residual mapping and y is the block output. This identity shortcut allows gradients to propagate more directly through the network during backpropagation, enabling stable training at deep layers.

AQI labeled sky image datasets are much smaller than large scale natural image datasets like ImageNet. Because of this, all convolutional layers of the ResNet50 backbone are kept frozen during training. This means their weights are not updated. This is a common transfer learning strategy. It lets the network reuse visual features that ResNet50 already learned from ImageNet, such as edges, textures, haze patterns, and color changes. Only the new classification layers added later are trained. Freezing the backbone also lowers the computation and memory needed for training, which fits the lightweight goal of this study. For an input image of size $224 \times 224 \times 3$, the backbone outputs a feature map of size $7 \times 7 \times 2048$. This means 2048 feature channels spread over a small 7×7 grid.

2) *Channel Reduction and Bilinear Upsampling*: The high channel dimensionality (2048) of the ResNet50 output is computationally expensive to process directly in subsequent stages. A 1×1 convolution is therefore applied to project the feature map from 2048 to 256 channels. Since a 1×1 kernel operates independently at each spatial location, this operation performs a learned linear combination across channels without mixing spatial information:

$$y_{i,j,k} = \sum_{c=1}^{2048} w_{k,c} \cdot x_{i,j,c} + b_k \quad (2)$$

where $x_{i,j,c}$ is the input activation at spatial location (i, j) and channel c , $w_{k,c}$ are the learned weights for output channel k , and b_k is the corresponding bias term. This yields a reduced feature map of size $7 \times 7 \times 256$, retaining the most salient channel information at a fraction of the computational cost.

The spatial resolution is then expanded from 7×7 to 56×56 using bilinear upsampling, so the feature map can later be split into equal patches. For each new pixel location, bilinear interpolation calculates its value as a weighted sum of the 4 nearest pixels in the smaller feature map:

$$\mathbf{F}_{\text{up}}(x', y') = \sum_{i=1}^2 \sum_{j=1}^2 w_{ij} \cdot \mathbf{F}_{\text{red}}(x_i, y_j) \quad (3)$$

Here, w_{ij} is the weight given to each neighboring pixel based on how close it is to the target location. This produces a smooth upsampled feature map of size $56 \times 56 \times 256$. The size 56 is chosen on purpose, since it divides evenly by 4. This allows the next step, patchification, to split the map into equal patches with no leftover pixels. This produces a smoothly interpolated feature map of size $56 \times 56 \times 256$. The target resolution of 56 is not arbitrary, it is chosen specifically because 56 is evenly divisible by 4, allowing the following patchification stage to partition the map into uniform, non-overlapping regions without padding or cropping artifacts.

3) *Spatial Patchification*: To preserve localized spatial information that would otherwise be lost under a single global pooling operation, the $56 \times 56 \times 256$ feature map is partitioned into a 4×4 grid of non-overlapping sub-patches. Each patch $P_{i,j}$ is defined as a contiguous $14 \times 14 \times 256$ crop of the feature map:

$$P_{i,j} = X[14i : 14(i+1), 14j : 14(j+1), :], \quad \text{for } i, j \in \{0, 1, 2, 3\} \quad (4)$$

where X is the $56 \times 56 \times 256$ feature map. This operation yields 16 independent sub-patches, each of dimension $14 \times 14 \times 256$, collectively covering the full spatial extent of the original image without overlap. Patchification allows the network to reason about localized regions of the sky image.

4) *Cross Stage Partial Block (CSPBlock) Processing*: Each of the 16 patches is processed independently through a shared CSPBlock, meaning the same set of weights is applied to every patch, rather than learning 16 separate sub-networks. This weight-sharing keeps the parameter count low and enforces consistent feature processing across all spatial regions, which is central to the lightweight design goal of this study.

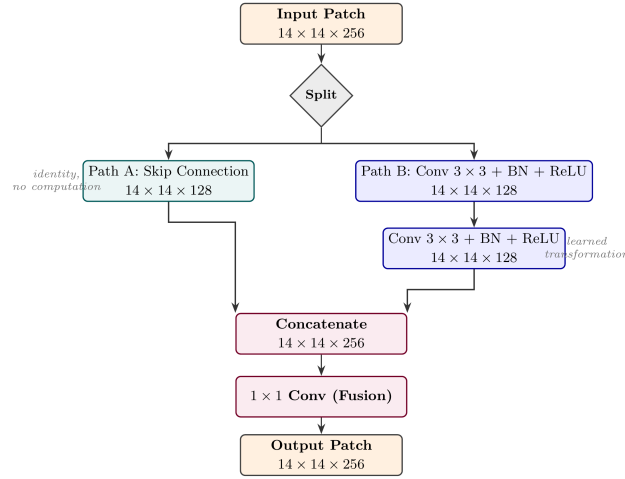


Fig. 7. Internal structure of the shared CSPBlock. The input patch is split into two 128-channel paths: an identity skip connection and a convolutional path (two 3×3 Conv + BN + ReLU stages), which are concatenated and fused via a 1×1 convolution.

The CSPBlock architecture is based on the Cross Stage Partial Network design [22], originally proposed to reduce redundant gradient information that arises when a feature map is reused across densely connected layers. Instead of passing the full channel volume through a convolutional stack, CSPNet partitions the input channels into two paths and merges them after independent processing. For each patch $P_{i,j}$ of shape $14 \times 14 \times 256$, the 256 channels are split evenly into two 128-channel paths:

$$P_{i,j} = [P_{\text{skip}} \parallel P_{\text{conv}}], \quad \text{where } P_{\text{skip}}, P_{\text{conv}} \in \mathbb{R}^{14 \times 14 \times 128} \quad (5)$$

The skip path P_{skip} is carried forward unchanged, preserving the original feature information with no additional computation. The conv path P_{conv} is passed through two successive 3×3 convolutional layers, each followed by batch normalization and ReLU activation:

$$P'_{\text{conv}} = \text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(P_{\text{conv}})))))) \quad (6)$$

Each convolution in the conv path is followed by batch normalization and a ReLU activation, which helps stabilize training and adds non-linearity. The two paths are then concatenated along the channel dimension and blended using a 1×1 convolution, following the same channel-mixing operation described in Equation (2):

$$P_{\text{out}} = \text{Conv}_{1 \times 1}([P_{\text{skip}} \parallel P'_{\text{conv}}]) \quad (7)$$

This split-transform-merge structure allows the network to retain a direct, unmodified copy of the original patch features while still learning new, non-linear transformations, without duplicating gradient flow through the entire channel volume. In this architecture, CSPBlock serves as a lightweight local feature refiner applied uniformly across all 16 spatial regions before aggregation.

5) *Patch Feature Aggregation via Global Average Pooling*: Following CSPBlock processing, each $14 \times 14 \times 256$ patch is reduced to a single 256-dimensional vector using Global Average Pooling (GAP), which computes the spatial mean of each channel independently:

$$v_{i,j,k} = \frac{1}{14 \times 14} \sum_{m=1}^{14} \sum_{n=1}^{14} P_{\text{out}}(m, n, k) \quad (8)$$

This is performed for every one of the 16 patches, producing 16 vectors $v_{i,j} \in \mathbb{R}^{256}$. These vectors are then averaged into a single image-level representation:

$$v_{\text{final}} = \frac{1}{16} \sum_{i,j} v_{i,j} \quad (9)$$

This two-stage pooling, first within each patch, then across patches, is a deliberate design choice rather than a simplification. Applying GAP once directly to the full $56 \times 56 \times 256$ map would collapse all spatial detail in a

single step, discarding any distinction between, for example, a hazy region in one part of the sky and a clear region in another. By pooling within each patch first, the network retains 16 locally-averaged descriptors before combining them, allowing patch-level variation to influence the final representation before it is summarized into a single vector.

6) *Classification Head*: The final 256 dimensional image representation v_{final} is passed through a small classification head. First, a fully connected layer projects it from 256 to 512 dimensions. This is followed by a ReLU activation, which adds non-linearity by zeroing out negative values:

$$\text{ReLU}(x) = \max(0, x) \quad (10)$$

To reduce overfitting, since the frozen backbone leaves only a small number of trainable parameters to fit the AQI dataset, dropout is applied with a probability of $p = 0.4$. This randomly turns off 40% of the activations during each training step, which forces the network to not rely too heavily on any single feature. Finally, a linear output layer maps the 512 dimensional representation down to the 6 AQI severity classes, producing the final class scores $\hat{y}$.

7) *Loss Function and Optimization*: The network is trained end to end, excluding the frozen backbone, using Cross-Entropy Loss. This loss combines a softmax normalization of the output scores with the negative log likelihood of the correct class. For an output vector $\hat{y}$ and true class label c , the loss for a single sample is:

$$L = -\log \left(\frac{\exp(\hat{y}_c)}{\sum_{j=1}^6 \exp(\hat{y}_j)} \right) \quad (11)$$

This loss is minimized using the Adam optimizer, which adjusts the learning rate for each parameter based on past gradients. Since ResNet50 is frozen, only the new layers are trained, using an initial learning rate of $\alpha = 0.0001$. A ReduceLROnPlateau scheduler monitors validation loss and halves the learning rate after 3 stagnant epochs, allowing smaller and more careful updates as training progresses.

III. EXPERIMENTAL RESULTS AND DISCUSSION

This section presents the empirical evaluation of the proposed framework for multi-class Air Quality Index (AQI) classification.

A. Evaluation Metrics

Model performance is assessed using four standard classification metrics: precision, recall, F1-score, and overall accuracy. For a given class, precision measures how many of the samples predicted as that class were actually correct, while recall measures how many of the true samples of that class were correctly identified:

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN} \quad (12)$$

where TP , FP , and FN denote true positives, false positives, and false negatives for a given class. F1-score is the harmonic mean of precision and recall, providing a single balanced measure of per-class performance:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (13)$$

Overall accuracy is reported as the ratio of correctly classified samples to the total number of test samples:

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \quad (14)$$

In addition to per-class metrics, two aggregate scores are reported: the macro-averaged F1-score, which treats all six classes equally regardless of their sample count, and the weighted-averaged F1-score, which weights each class by its number of test samples. Both are reported to give a complete picture of performance across the class-imbalanced test set.

To visualize class-wise prediction behavior, a confusion matrix is also used, where rows represent the true class and columns represent the predicted class. The normalized confusion matrix expresses each row as a percentage of that class's total samples:

$$\text{CM}_{\text{norm}}(i, j) = \frac{\text{CM}(i, j)}{\sum_k \text{CM}(i, k)} \times 100 \quad (15)$$

allowing classes with different sample counts to be compared fairly.

B. Training Dynamics and Convergence

The training and validation progression across epochs was evaluated to assess model learning stability:

- **Loss Reduction:** The training and validation loss curves, computed using the Cross-Entropy Loss defined in the Methodology section, decrease smoothly over epochs without divergence, indicating stable optimization and effective regularization.
- **Accuracy Growth:** Validation accuracy exceeds 90% within the initial epochs and stabilizes near the final test accuracy of 96.73% as training completes. Notably, validation accuracy exceeds training accuracy through much of the training process, consistent with the regularizing effect of dropout, which is active only during training and temporarily suppresses the model's training-time performance.

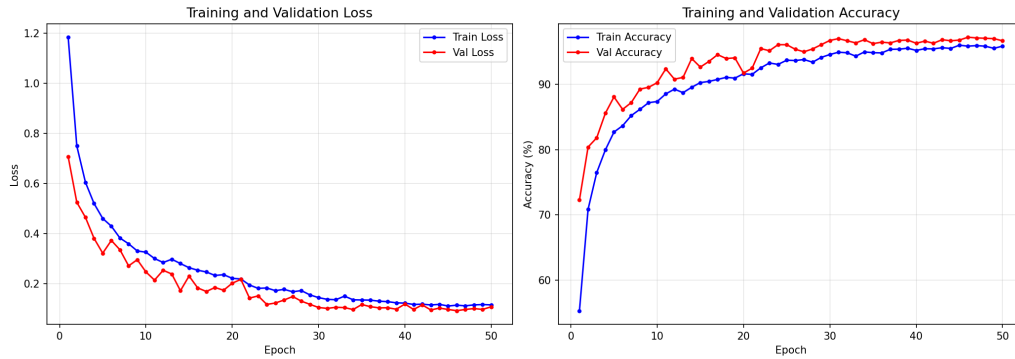


Fig. 8. Training and Validation Loss and Accuracy curves across the training run, showing smooth convergence without overfitting.

C. Overall and Class-Wise Classification Performance

On the full test set of 1836 images, the model correctly classified 1776 samples, corresponding to an overall accuracy of **96.73%**. The macro-averaged F1-score was **0.9666** and the weighted-averaged F1-score was **0.9673**, indicating consistent performance across classes regardless of their sample frequency.

To evaluate model reliability across distinct pollution levels, class-wise precision, recall, and F1-score were extracted alongside raw and normalized confusion matrices.

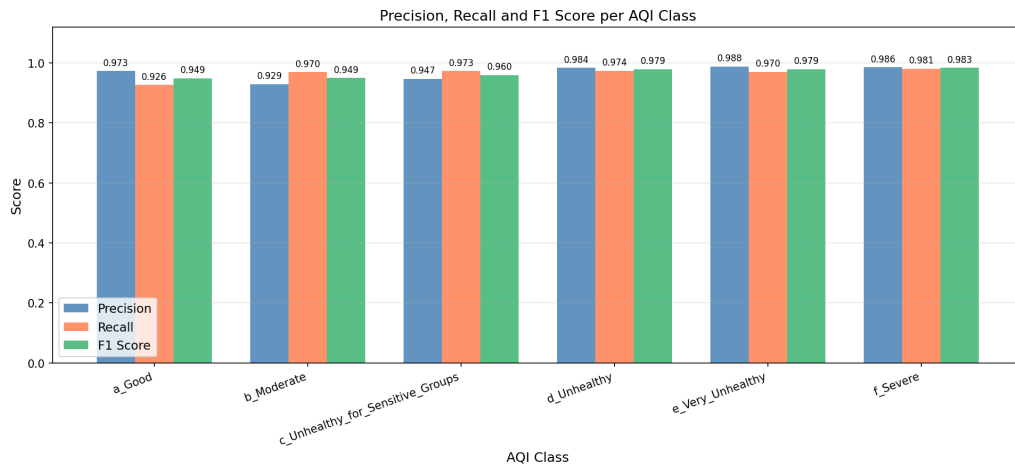


Fig. 9. Precision, Recall, and F1-Score metrics across the 6 target AQI severity classes.

- **Severe Pollution Detection:** The Severe (f_Severe) class recorded the highest F1-score of 0.983, with a recall of 0.981, corresponding to 207 out of 211 samples correctly identified.
- **Unhealthy AQI Classes:** Performance across the intermediate unhealthy classes remained consistently high:
 - e_Very_Unhealthy: F1-score of 0.979 (recall 0.970, 327 of 337 correctly identified).

- d_Unhealthy: F1-score of 0.979 (recall 0.974, 375 of 385 correctly identified).
- c_Unhealthy_for_Sensitive_Groups: F1-score of 0.960 (recall 0.973, 429 of 441 correctly identified).
- **Good AQI Class:** The a_Good class achieved an F1-score of 0.949, with a high precision of 0.973 but a comparatively lower recall of 0.926 (214 of 231 correctly identified). This gap is driven by 13 true a_Good images being misclassified as b_Moderate, suggesting the model occasionally under-detects Good conditions when visual haze levels are borderline between the two categories.
- **Moderate AQI Class:** The b_Moderate class recorded the lowest precision among all six classes at 0.929, despite a strong recall of 0.970 (224 of 231 correctly identified). This is largely explained by the same 13 misclassified a_Good samples landing in the Moderate prediction column, indicating that Good and Moderate remain the two most visually overlapping classes for the model, with confusion occurring in both directions.

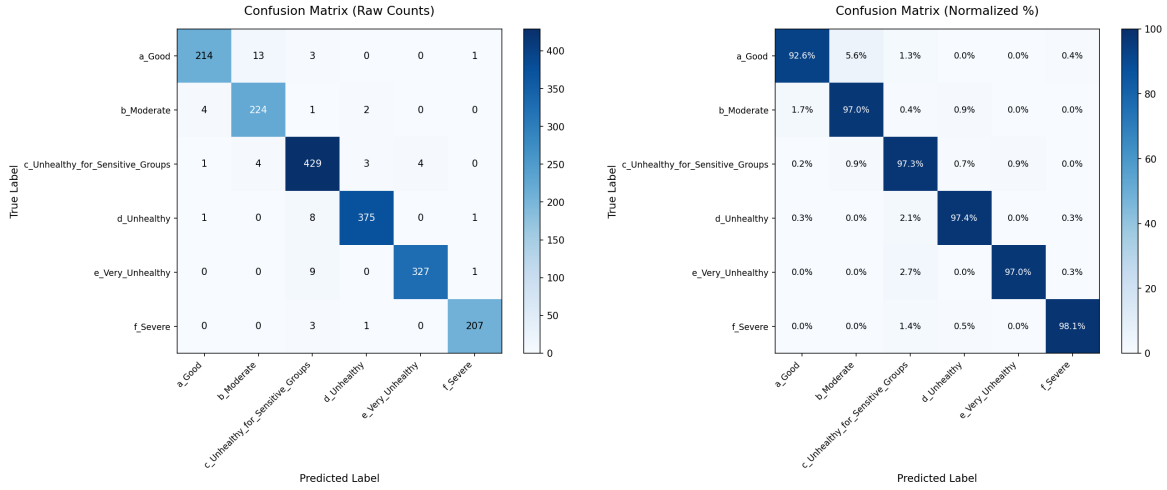


Fig. 10. Confusion matrices showing raw sample counts (left) and normalized percentage accuracy (right) across all AQI classes.

D. Computational Efficiency and Inference Speed

To demonstrate deployment feasibility for real-time edge computing and mobile devices, the proposed framework was benchmarked against standard deep learning backbones, including VGG16, VGG19, InceptionV3, MobileNetV2, and a ResNet50 baseline. Inference time was measured as the average time per image over 3 timed runs, following a 10-batch GPU warm-up phase to exclude one-time initialization overhead. All measurements were performed using a batch size of 32 on an NVIDIA Tesla T4 GPU, with GPU operations explicitly synchronized before and after timing to ensure accurate measurement.

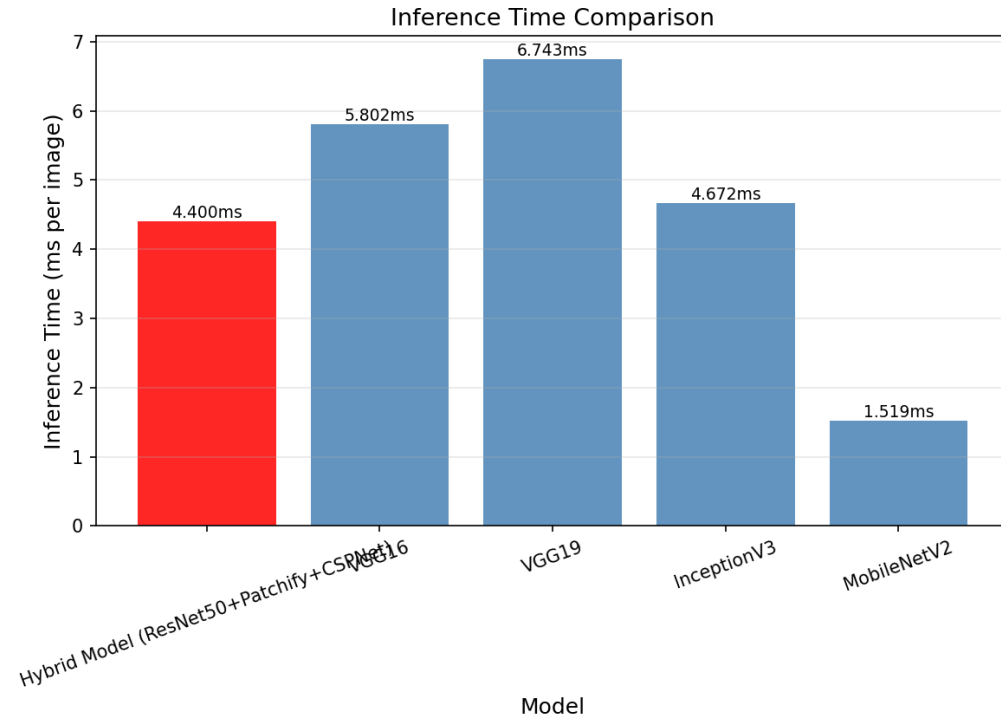


Fig. 11. Inference time comparison (ms per image) of the proposed the proposed model against standard deep learning backbones.

- **Inference Speed Analysis:** The proposed framework achieves an average execution time of 4.400 ms per image, faster than VGG16 (5.802 ms), VGG19 (6.743 ms), and InceptionV3 (4.672 ms), while remaining moderately slower than the lighter ResNet50 baseline (3.456 ms) and MobileNetV2 (1.519 ms).
- **Accuracy vs. Latency Trade-Off:** Although MobileNetV2 offers the fastest runtime (1.519 ms), its test accuracy is substantially lower at 74.73%. Similarly, the ResNet50 baseline, VGG16, VGG19, and InceptionV3 all fail to exceed 79% accuracy despite comparable or higher latency, indicating that raw backbone depth alone does not translate into strong AQI classification performance on this task.
- **Optimal Balance:** The proposed framework achieves the highest accuracy of all evaluated models at 96.73%, roughly 18–28 percentage points above every baseline, while maintaining inference latency within the same range as the standard backbones. This places the proposed model in the optimal top-left region of the accuracy-latency trade-off plot, demonstrating that the proposed patch-based CSP design achieves substantially higher accuracy without incurring the heavier computational cost typically associated with such gains.

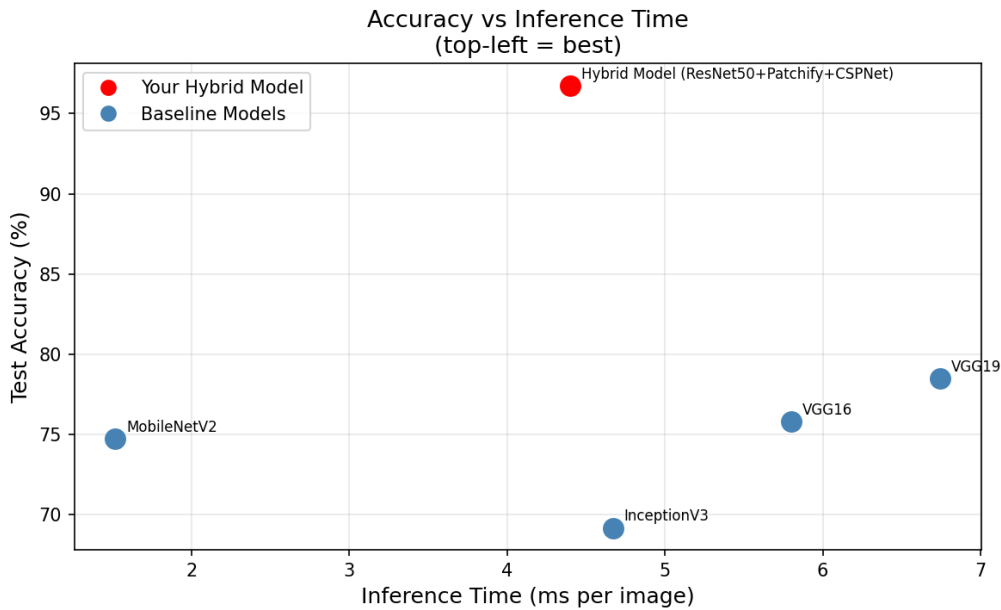


Fig. 12. Accuracy vs. Inference Time trade-off plot, highlighting the optimal top-left performance of the proposed model.

E. Comparison with Existing Literature

To evaluate the effectiveness of the proposed framework, its performance was benchmarked against literature across traditional processing, classical ML, single-modality DL, and multimodal frameworks, as detailed in Table IV. It should be noted that these comparisons are not fully controlled: reported accuracies come from studies with differing class granularity (3-class versus 6-class severity levels), input modalities (image-only versus multimodal image and sensor fusion), and dataset compositions, and should therefore be interpreted as an indication of relative performance rather than a strictly like-for-like benchmark.

TABLE IV
COMPARISON OF PROPOSED METHOD WITH LITERATURE REVIEW APPROACHES

Ref	Approach / Architecture	Modality	Accuracy (%)
[11]	Luminance/Histogram + KNN	Image Only	92.00%
[13]	Color Moments + SVM	Image Only	82.30%
[14]	Texture Analysis + ML	Image Only	86.00% / 80.00%
[16]	Multi-scale Dilated CNN	Image Only	94.20%
[17]	AQFusionNet (EfficientNet-B0)	Multimodal	92.02%
[15]	ResNet50 Backbone	Image Only (3 Classes)	97.87%
[18]	Vision-AQ (ResNet50 + MLP)	Multimodal	99.00%
[19]	CapsuleNet Framework	Multimodal	99.98%
Prop.	Proposed Framework	Image Only (6 Classes)	96.73%

- **Superiority over Traditional & ML Methods:** The proposed framework outperforms classical approaches such as KNN [11] (92.0%), SVM [13] (82.3%), and texture classifiers [14] (80.0%–86.0%), as well as the multi-scale dilated CNN of [16] (94.2%), despite operating on a finer-grained 6-class problem rather than a coarser 2–3 class split typically used in these approaches.
- **Competitive Edge against Deep Learning Models:** While multimodal architectures such as Vision-AQ [18] (99.0%) and CapsuleNet [19] (99.98%) achieve higher accuracy, they rely on auxiliary tabular sensor data in addition to images. the proposed model achieves 96.73% accuracy using visual sky images alone, outperforming other image-only and lightweight multimodal baselines [16], [17], and remains competitive with the 3-class ResNet50 baseline of [15] (97.87%) despite solving a substantially harder 6-class classification task.

IV. CONCLUSION AND FUTURE WORK

A. Conclusion

This study introduced a **lightweight hybrid deep learning framework** for automated AQI classification using ground-level sky imagery. By combining a frozen ResNet50 backbone with 1×1 reduction, bilinear upsampling, spatial patchification, and a shared CSPBlock, the network extracts global atmospheric patterns and localized haze textures effectively. Evaluated on 12,240 images across 6 AQI categories, the model achieves 96.73% accuracy with an inference speed of 4.400 ms per image on a single NVIDIA Tesla T4 GPU, offering a favorable accuracy-latency trade-off relative to standard backbones of comparable or higher computational cost, and making it well suited for edge and mobile deployment. Performance was slightly weaker on the Moderate AQI class, reflecting the visual similarity between adjacent severity levels.

B. Future Work

Future research directions include:

- **Real-Time Edge Deployment:** Model compression via quantization and pruning for deployment on resource-constrained micro-controllers and IoT platforms.
- **Multimodal Data Fusion:** Incorporating meteorological sensor data (humidity, temperature, wind speed) to resolve ambiguous boundary cases.
- **Dataset Expansion:** Expanding geographic and seasonal data coverage, including nighttime low-light conditions.

REFERENCES

- [1] World Health Organization. (2023) Ambient (outdoor) air pollution. [Online]. Available: <https://www.who.int/health-topics/air-pollution>
- [2] M. S. Islam, "Air pollution induces respiratory diseases in bangladesh: A perception based study," *IJFMR*, vol. 7, no. 2, pp. 1–10, 2025.
- [3] American Lung Association. (2026) Air quality index. [Online]. Available: <https://www.lung.org/clean-air/outdoors/air-quality-index>
- [4] Y. Farhaoui, S. S. Alaoui, and B. Aksasse, "Air pollution prediction through internet of things technology and big data analytics," *Int. J. Comput. Intell. Stud.*, vol. 8, no. 3, p. 177, 2019.
- [5] World Health Organization, *WHO Global Air Quality Guidelines: Particulate Matter (PM_{2.5} and PM₁₀), Ozone, Nitrogen Dioxide, Sulfur Dioxide and Carbon Monoxide*. Geneva: World Health Organization, 2021. [Online]. Available: <https://iris.who.int/handle/10665/345329>
- [6] R. M. Harrison *et al.*, "Health risk associated with airborne particulate matter and its components in jeddah, saudi arabia," *Science of The Total Environment*, vol. 590–591, pp. 531–539, Jul. 2017.
- [7] J. H. Choi, J. S. Kim, Y. C. Kim, Y. S. Kim, N. H. Chung, and M. H. Cho, "Comparative study of pm_{2.5} - and pm₁₀ - induced oxidative stress in rat lung epithelial cells," *J. Vet. Sci.*, vol. 5, no. 1, p. 11, 2004.
- [8] M. K. Hossain, R. I. Zerín, H. Rashid, U. S. P. Kakon, M. S. S. Trina, L. A. Eva, S. Ali, M. O. F. Molla, M. S. Jahan, M. A. Seddikiy, S. Kumar, M. Maniruzzaman, G. R. MacFarlane, and R. Islam, "Spatial distribution and AirQ+-based health risk analysis of particulate matter in Bangladesh's industrial hubs," *Water, Air, & Soil Pollution*, vol. 237, no. 11, p. 660, jun 2026. [Online]. Available: <https://doi.org/10.1007/s11270-026-09294-7>
- [9] World Health Organization. (2021) Who global air quality guidelines: Questions and answers. [Online]. Available: <https://www.who.int/news-room/questions-and-answers/item/who-global-air-quality-guidelines>
- [10] X. Xie, I. Semanjski, S. Gautama *et al.*, "A review of urban air pollution monitoring and exposure assessment methods," *ISPRS Int. J. Geo-Inf.*, vol. 6, no. 12, p. 389, Dec. 2017.
- [11] S. Hao, P. Wang, and Y. Hu, "Haze image recognition based on brightness optimization feedback and color correction," *Information*, vol. 10, no. 2, p. 81, Feb. 2019.
- [12] Y. Chu, F. Chen, H. Fu, and H. Yu, "Detection of air pollution in urban areas using monitoring images," *Atmosphere*, vol. 14, no. 5, p. 772, Apr. 2023.
- [13] W. Wang, W. Shen, B. Chen, R. Zhu, and Y. Sun, "Air quality index forecasting based on svm and moments," in *Proc. 2018 5th Int. Conf. Systems and Informatics (ICSAI)*, Nanjing, IEEE, Nov. 2018, pp. 851–855.
- [14] M. Chuchro, W. Sarlej, M. Grzegorzczak, and K. Nurzyńska, "Application of photo texture analysis and weather data in assessment of air quality in terms of airborne pm₁₀ and pm_{2.5} particulate matter," *Sensors*, vol. 21, no. 16, p. 5483, Aug. 2021.
- [15] R. K. Kaushal, H. Pandey, G. Kalaiarasi, S. Arya, D. Rawat, and N. L., "Evaluation of deep learning approaches for air quality analysis using an image dataset," in *Proc. 2024 2nd Int. Conf. Intelligent Cyber Physical Systems and Internet of Things (ICoICI)*, Coimbatore, India, IEEE, Aug. 2024, pp. 1378–1383.
- [16] Z. Wang, Y. Yang, and F. Wu, "A lightweight air quality monitoring method based on multiscale dilated convolutional neural network," *IEEE Trans. Ind. Inf.*, vol. 20, no. 12, pp. 14 184–14 192, Dec. 2024.
- [17] K. A. Kushal and A. A. Mamun, "Aqfusionnet: Multimodal deep learning for air quality index prediction with imagery and sensor data," *arXiv preprint arXiv:2509.00353*, 2025.

- [18] F. Mehmood, S. U. Rehman, and A. Choi, "Vision-aq: Explainable multi-modal deep learning for air pollution classification in smart cities," *Mathematics*, vol. 13, no. 18, p. 3017, Sep. 2025.
- [19] H. Karamti, A. Aldrees, N. S. Alghamdi, M. Umer, and M. Nappi, "Intelligent multi-modal data implementation with capsulenet for accurate air quality index prediction," *Complex Intell. Syst.*, vol. 11, no. 12, p. 488, Dec. 2025.
- [20] A. Rouniyar, "Air Pollution Image Dataset from India and Nepal," Kaggle, 2023, [Online]. Available: <https://www.kaggle.com/datasets/adarshrouniyar/air-pollution-image-dataset-from-india-and-nepal>. [Online]. Available: <https://www.kaggle.com/datasets/adarshrouniyar/air-pollution-image-dataset-from-india-and-nepal>
- [21] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016, pp. 770–778.
- [22] C.-Y. Wang, H.-Y. M. Liao, I.-H. Yeh, Y.-H. Wu, P.-Y. Chen, and J.-W. Hsieh, "Cspnet: A new backbone that can enhance learning capability of cnn," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, 2020, pp. 1571–1580.